Krishal Hriday

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SUMMARY

CS undergrad building reliable GenAI workflows – prompt engineering, LLM evaluation, observability, and Python automation. IEEE-published researcher; active open-source contributor.

EDUCATION

PES University Aug 2023 – May 2027
B.Tech, Computer Science and Engineering *Bengaluru, Karnataka*
Coursework: DSA, OOAD, DBMS, OS, Networks, Compiler Design, Cloud Computing, ML, Big Data, Data Analysis

SKILLS

Languages: Python, JavaScript, TypeScript, C++
Backend & Automation: FastAPI, Flask, REST APIs, Streamlit, Python scripting, Git/GitHub Actions (CI)
Cloud & Data: Docker, AWS S3, Apache Kafka, Apache Spark, Modal (serverless GPU)
GenAI & Prompting: Prompt Engineering & Versioning, RAG, LangChain, Hugging Face, PyTorch
Evals & Observability: Promptfoo, DeepEval, Langfuse, Golden Datasets, Red-Teaming, Guardrails, Regression Testing
Frontend & DBs: React, Next.js, Tailwind CSS; Node.js; PostgreSQL, MongoDB, SQLite, Prisma

ACHIEVEMENTS

- **IEEE Xplore Publication:** First-author paper on LLM tool-use reliability, accepted at IEEE ICWITE 2026 (see Projects).
- **Academic Excellence:** Awarded Distinction in the 5th and 6th semesters.

PROJECTS

Enterprise AI Operations Triage – GenAI workflow with evals & guardrails
- Built a triage prototype (FastAPI, Streamlit, Ollama) with retrieval, citation checks routing unsupported answers to human review, and supervisor approval; wrote reusable prompt templates and a runbook for non-technical users.
- Created an eval stack: 50-case golden set + 8 red-team cases as a Promptfoo CI gate, DeepEval scoring, and Langfuse tracing with prompt v1/v2 comparison.

Brittle Semantic Collapse Under Perturbation – IEEE ICWITE 2026
- Designed a 300+ query evaluation harness (clean, typo, ambiguity cases; syntactic + semantic validators) showing 3B LLMs drop from 90% to 15% semantic accuracy under minor perturbations; ran an 81-run prompt ablation (Welch's ANOVA $p < 10^{-83}$) proving failures are model-intrinsic, not prompt-driven.

AI Podcast Clipper – Full-stack GenAI SaaS
- Built an end-to-end SaaS (Next.js, FastAPI, Modal GPU) that transcribes podcasts (WhisperX), uses prompted Gemini to pick viral moments, and auto-renders vertical clips with speaker cropping and subtitles.
- Implemented Stripe credit billing, S3 storage, Auth.js login, and an Inngest job queue for long-running renders.

Distributed Image Processing Pipeline (Apache Kafka)
- Engineered a fault-tolerant, tile-parallel image pipeline on Kafka with manual partitioning for guaranteed load balancing, async reassembly, and a real-time Flask monitoring dashboard.

OPEN SOURCE CONTRIBUTIONS

bernstein – deterministic orchestrator for CLI coding agents (beta)
- PR #3678: Added auditability/guardrail logging – recorded automatic worker-kill decisions into a signed HMAC audit chain as `stall.verdict` events (heartbeat/stall detector, evidence, threshold), replacing an in-memory dict lost on daemon restart.
- PR #3721: Wired signed `escalation.receipt` evidence into all 3 auto-kill paths, giving each kill a reconstructable, offline-verifiable failure window on the audit chain.